



BACHELOR IN SOFTWARE ENGINEERING (HONOURS)

FINAL EXAMINATION JANUARY 2023

Course : EC3307 (Object and Component Technology) Time: 9.00am – 12.00noon
(3 hours)

Lecturer : Ms. Rajesvary Rajoo

Date: 11 May 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **3** printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Examine the following code:

```
public abstract class Animal {  
    String name;  
    String description;  
    String covering;  
    AttackBehaviour myAttack;  
  
    public Animal (String name) {  
        this.name = name;  
    }  
    abstract void move();  
    void performAttack() {myAttack.attack();}  
}
```

- i. Assume that different types of animals (Eagle, Snake and Spider) attack in a different way (SwoopAttack, PoisonousBiteAttack and SpitAttack). Determine the most suitable design pattern to adapt the above code and to accommodate these distinct attack behaviors. Explain the rationale behind the choice. **(4 MARKS)**
 - ii. Draw the class diagram for the design pattern that you have determined in a(i). **(6 MARKS)**
 - iii. Describe the design principle that this pattern adheres to. **(5 MARKS)**
- b. The Singleton Pattern relies on two constructs or features to ensure that: (1) at most only one object of a class is ever created by clients, and (2) a reference to the unique instance of the singleton class is always accessible to clients.
- i. Explain how these could be implemented in programming? **(4 MARKS)**
 - ii. Evaluate the effectiveness of the implementation of these features. **(6 MARKS)**

- 2 a. What is the difference between the specification of a method and its implementation? **(4 MARKS)**
- b. A student sitting an exam gives the following subtly wrong definitions for pre- and post-conditions:
"A precondition of a method is something that is always true whenever the method is entered.
A post-condition is something that is always true when the method is completed."
Explain why both definitions are wrong. **(4 MARKS)**
- c. The java.util.List interface has a get (int index) method which returns the element at the specified position in this list. Write pre- and post-conditions in English for this method using the variables index and result, to indicate the value returned **(4 MARKS)**
- d. Explain why preconditions cannot be **AND**ed together just like postconditions and object invariants in inheritance when subclass inherits specifications from the superclass. **(6 MARKS)**
- e. Explain by using a suitable example why defensive programming is not encouraged to be used in programming. Suggest an alternative. **(7 MARKS)**
3. a. Differentiate upcasting and downcasting in Java. Explain when each should be used. **(6 MARKS)**
- b. Explain **ONE (1)** advantage of implementing generic parameter (type parameter) in programming. Illustrate your answer using an example. **(5 MARKS)**
- c. Explain what the following code means for the object shapeList **(4 MARKS)**

ArrayList<Shape> shapeList = new ArrayList<Shape>
where Shape is a class.

- d. Write a generic class Stack<T> that can be used to represent stacks of objects of type T. The class should include methods push(), pop(), and isEmpty(). Use an ArrayList to hold the items on the stack inside the class. **(10 MARKS)**
- 4 a. Distinguish how the implementation of the switch statement in C# differs from Java with regards to the use of the goto statement. **(5 MARKS)**
- b. Outline **ONE (1)** advantage of the Common Type System (CTS) of the .NET platform. **(5 MARKS)**
- c. Contrast the classic Java approach of the Java Virtual Machine with the .NET way of using an intermediate language. **(5 MARKS)**
- d. Differentiate software component and component software. **(5 MARKS)**
- e. Argue **ONE (1)** of the deciding factors to determine whether you would choose Java technology or .Net technology as a platform for application development. **(5 MARKS)**

-END OF QUESTION PAPER-